BELLO, Malik Pelumi

AI/ML Engineer

AI/ML engineer who takes models from data to live production in banking, fraud, health and African languages. At Wema Bank I build agentic AI, graph-neural-network customer models, SME lending models and fraud-model features that run in live systems. Beyond work I led PHC Scribe, a fine-tuned 8B Nigerian-language LLM and speech system for clinical documentation, with a live API.

belloayopelumi@gmail.com | 08146234068 | Nigeria | malikbello.is-a.dev | Tech Socials

WORK EXPERIENCE

- **AI/ML Engineer (since 02/2026); previously Data Scientist**
  Wema Bank Plc, Nigeria — 03/2025 – Present
  - Built **MSIP** (Mystery Shopping Intelligence Platform), an agentic AI system with harness engineering for the Branch Experience Management team, cutting 1–3 months of work to one day.
  - Built an adaptive customer-intelligence model: an ensemble of graph neural networks learns customer behavioural embeddings that feed temporal propensity and recommendation models. Delivered end to end on Microsoft Fabric, with outputs integrated directly into the live MoEngage system and banking apps.
  - Delivered end-to-end SME lending models under a champion-challenger framework (MLflow, Fabric MLOps), with full A/B testing and feedback loops.
  - Contributed to the enterprise fraud-detection model: built the feature stores, then evaluated and re-tuned the champion model, which helps catch about 80% of confirmed fraud while cutting manual review by about 60% (internal evaluation).
  - Engineered agentic AI systems with federated knowledge-base access, tool calling and SQL generation, plus RAG agents for HR and Legal; built ETL-to-EDA pipelines for API data and logs that saved millions of naira.
- **Team Lead and Engineer: PHC Scribe and Tákàdá (health AI)**
  NAIC 2026 · github.com/malikbello/phc-scribe — 2026
  - Led a multidisciplinary team of engineers, nurses and a pharmacist building a system that turns a spoken antenatal visit in English, Pidgin or Yorùbá into a structured record with WHO-based danger-sign flags.
  - Fine-tuned N-ATLaS 8B with QLoRA (Q4 GGUF) and N-ATLaS speech models (int8) for Yorùbá, Pidgin and Nigerian English; published the models on Hugging Face.
  - Added a safety layer: every finding must quote the words it came from, and polarity and support checks flag negated or unsupported findings for the nurse.
  - Benchmarked on frozen test sets never used to choose settings; shipped a rate-limited FastAPI service on a Modal GPU, an offline clinic hub (Docker, llama.cpp, no GPU) and a React phone app with CI-gated pull requests. Live demo: takada-app.vercel.app
- **Data Analyst**
  ICT Directorate, Olabisi Onabanjo University — 5/2024 – 2/2025
  - Performed ETL on the students' database using Excel, Microsoft Azure and on-prem MySQL; analysed CBT examination schedules, attendance and sessional payment data.
- **Data Science Intern**
  3MTT Nigeria || The Founder's Hub Network — 02/2024 – 12/2024
  - Built ESAAM (Exploratory and Sensitivity Analysis App), a LightGBM app predicting the Multidimensional Poverty Index and the impact of a tech training on it; also modelled cancer prediction with logistic regression on the Wisconsin Health Dataset. [Report, Repo, Video]
- **Data Analysis Intern**
  HNG Internship 11 — 07/2024 – 08/2024
  - Designed a dimensional sales model in SQL; modelled productivity in Nigeria from economic metrics; predicted flood events from historical flood, weather and sea data; used geospatial analysis to detect rigging outliers in election data.
- **Tax Analysis Intern**
  KPMG US Forage Tax Simulation — 08/2024
  - Researched and made tax recommendations, modelled and cleaned data in Excel, and wrote an IRAC research memorandum for the client.
- **Freelance Data & Research Analyst, Digital Marketer, and CopyWriter**
  Self-employed — 01/2020 – 12/2024
  - Scraped thousands of online leads and feedback for decision-making; built linear and logistic models for clients in finance, healthcare and education; designed quasi-experimental studies.
  - Built landing pages, sales funnels and automations with sizeable ROI increase; ran digital marketing for cryptocurrency, audit and real-estate clients.
- **Graduate Research Assistant and Analyst**
  Obafemi Awolowo University (OAU) — 07/2023 – 01/2024
  - Designed a healthcare community programme for under-5s using the Common Risk Factor Approach; analysed primary and secondary data, tested hypotheses with parametric and non-parametric methods, wrote manuscripts and designed 4 poster presentations.
- **Head of Technical and Supplementary Services**
  Lavender Tutors — 03/2020 – 03/2021
  - Coordinated computing resources, blueprinted a web app for tracking learning outcomes and tutor efficiency, and designed an income-payment-benefit plan.

EDUCATION

M.Sc. Financial Engineering — 01/2025 – 01/2027
WorldQuant University (USA)

B.Sc.Ed Biology (First Class; 4.83/5.00) — 04/2017 – 07/2023
Obafemi Awolowo University (OAU)

Professional Diploma in Data Analytics and Visualisation — 03/2024 – 02/2025
Baze University, Abuja

TECHNICAL SKILLS

ML and AI
Python, deep learning, graph neural networks, LightGBM/XGBoost, QLoRA fine-tuning, RAG, agents and tool calling, speech recognition

MLOps
MLflow, Microsoft Fabric, feature stores, champion-challenger and A/B testing, Docker, GitHub Actions CI, GGUF/llama.cpp

Data and Engineering
SQL, MySQL, FastAPI, ETL, feature engineering, out-of-time validation, Azure AI, AWS, Git

Data Analysis
SPSS, Excel, Power BI, hypothesis testing, report writing, MS Office, CanvaPro

PROFESSIONAL CERTIFICATES (IONS)

- Microsoft Certified: Azure AI Engineer Fundamentals (AI-900)
- Certified Data Science Professional and Scholar (CSDP || IIIPS)
- AWS AI Engineer (Udacity Nanodegree)
- IDR Statistical Consulting Associate
- HP Life Data Science and Data Analytics Certificate
- Microsoft and LinkedIn Professional Career Essentials in Data Analysis
- GitHub Career Essentials Professional Certificate
- Microsoft Data Modelling in Power BI
- Alx AI Career Essentials (AiCE)
- Cisco English for IT 1 & 2

VOLUNTEERING EXPERIENCE

- **Research Lead**
  Instride Pathways Youth Foundation — 05/2024 – Present
  - Leading a team comparing Logistic Regression, SVM and XGBoost on health datasets; gathered data on private secondary schools across Nigeria for CIAY; moderator and interim secretary.
- **Member, Humanity First Foundation** — 2023 – Present
  - "Pad-Up-A-Girl" project (11/23, 01/24): over 1000 students in Oyo and Osun received sanitary pads and menstrual-hygiene sensitisation.

PUBLICATIONS

- **Bello, M.P.**, Kareem, A. O., & Bamidele, E. F. (2025). Interactivity as a Retention Factor for Learning Biology Through the Protégé Effect. *Journal of Teaching and Learning*, 19(1), 107–130 {Scopus-rated Journal}
- Kareem, A. O., Oniyide, Z. & **Bello, M. P.** (2024). The Influence of Emigration Potential (Japa Syndrome) on the Teaching Motivation and Teaching Efficacy of Pre-Service Teachers. *SFJESGS*, 6(3), 235–251.

RESEARCH CONFERENCE

- Oral and Technical Presentation, 12th International Mardin Artuklu Scientific Researches Conference, Türkiye. 17–19/08/2024

SCHOLARSHIPS AND AWARDS

- Won Kaggle Bronze Award for Google Global AI Capstone Competition — 2025
- Earned an OAU Star Award of Excellence — 2024
- Awarded best debater, MTG NYSC Rivers State Inter-platoon Debate — 2024
- Summa cum laude; Chief Dr (Mrs) Oluremi Tinubu Prize for Best Graduating Student in STE Biology; Babs Fafunwa Educational Foundation Award — 21/22 Set

GENERAL SKILLS

Time Management | Team Work | Resourcefulness | Effective Communication | Leadership | Patience